When It Gets to Court: An Analysis of Candidates and Voters in a Judicialized Election

Evidence from Brazilian Municipal Elections, 2020–2024

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Does Judicial Intervention Reshape the Electoral Arena?

Research question:
Do pre-election adversarial lawsuits change the behaviour of *both* voters and candidates in Brazilian municipal elections?

Why this paper: the information degradation mechanism

- Pre-election lawsuits raise legal costs for candidates (**entry deterrence**) and generate ambiguous signals for voters (**withdrawal from candidate choice**)
- Voters cannot readily tell a meritorious challenge from strategic harassment — litigation adds noise, not signal
- Under Brazil's compulsory voting, voter disaffection surfaces as **blank ballots**, not abstention

Key Findings

Bartik shift-share IV · $F = 19.6$ · $N = 5{,}560$ municipalities

- **Blank vote rate** $\uparrow$ $+1.3$ pp ($p = 0.018$; survives tF correction) — voters withdraw from candidate choice
- **Candidate pool** contracts: -6.6% executive, -7.5% legislative — entry deterrence channel
- **Competition and turnout**: precise nulls after tF correction — no reallocation, no demobilization

Setting: Brazil, Municipalities, 2020–2024

- **Tribunal Superior Eleitoral (TSE)** + 27 *Tribunais Regionais Eleitorais* + 2,800+ first-instance electoral zones; adjudicates eligibility, campaign conduct, and propaganda violations *before* election day
- **5,570 simultaneous municipal elections** per cycle; directly elected mayor; uniform electoral rules under CF Art. 29
- **Treatment:** $\Delta \log(1 + \text{adversarial lawsuits}_m)$, 2020→2024; ∼1.2M adversarial cases in 2020, ∼1.1M in 2024
- **Identification:** 2020 lawsuit topic mix (shares) × state-level 2020→2024 topic growth (shifts) = Bartik IV
- **Analysis sample:** N = 5,560 municipalities; SE clustered by 2,187 principal electoral zones

Road Map

Related Literature and Contribution

Judicialization and Electoral Courts

- Tate & Vallinder (1995), Helmke & Staton (2011): judicialization as politics migrating into courts; Latin American waves driven by institutional diffusion
- Taylor (2008), Sadek (1995): TSE as constitutive political actor; Brazilian electoral justice literature

↪ *This paper: first causal estimate at the micro-electoral level — filings before election day, not cassação outcomes*

Voter Behavior and Compulsory Voting

- Power & Roberts (1995): institutional drivers of blank/spoiled votes in Brazil's compulsory system
- Driscoll et al., Castro (2025): blank votes as protest in low-information settings

↪ *This paper: first causal identification of judicial activity as a driver of blank votes*

Shift-Share Identification and Candidate Entry

- GPS (2020, AER), BHJ (2022, ReStud), Ash, Morelli & Vannoni (2025, JPE): shift-share IV design and compliance
- Besley & Coate (1997): entry deterrence; Ferraz & Finan (2008): information and accountability in Brazil

↪ *This paper: first Bartik IV applied to electoral litigation; full GPS/BHJ/AKM compliance with tF correction*

Three-tier judicial structure:

1. **Zona Eleitoral (ZE)** — first instance; 2,800+ nationwide
2. **Tribunal Regional Eleitoral (TRE)** — 27 state appellate courts
3. **Tribunal Superior Eleitoral (TSE)** — national apex

Pre-election jurisdiction:

- Candidate registration challenges (*impugnação*)
- Abuse of economic/political power
- Campaign finance violations
- Prohibited propaganda; vote-buying
- Conduct prohibitions (*conduta vedada*)

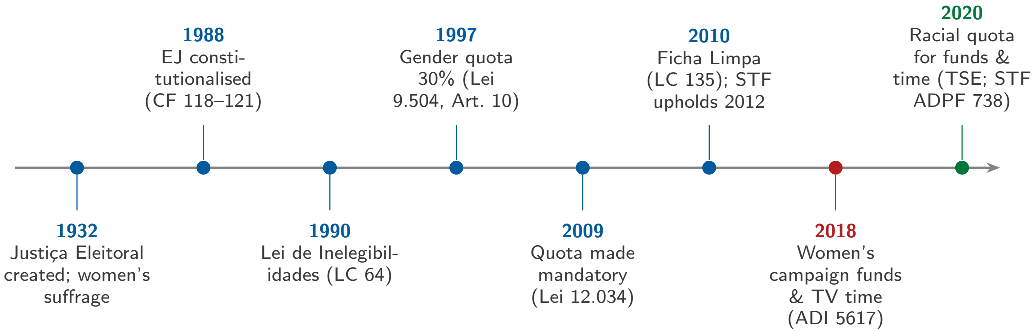
Election Timeline

1. Candidate registration (August)
2. **Pre-election lawsuits** (Aug–Oct)
3. **Election Day** (Oct/Nov)
4. Post-election proceedings
5. Cassation if ruling goes against winner

2020: Election day = November 15
2024: Election day = October 6

We use only cases filed *before* these dates.

The TSE as a Constitutive Actor — Electoral-Rule Milestones



Actor: statute (Congress), STF, TSE. Horizontal spacing not to scale. Sources: TSE; STF (ADI 5617; ADPF 738).

Zoom: 2020–2025 — The TSE Becomes a Central Political Actor

2020 — **Aug–Oct:** racial quota — FEFC & free TV time must be *proportional* to Black candidates (TSE, 25 Aug; STF ADPF 738 brings it forward to 2020).

2020 — **Nov:** municipal elections **postponed to November** (COVID-19; EC 107/2020). ← *paper's baseline cycle.*

2022 — TSE assumes disinformation/content-removal powers (Res. 23.714/2022); Moraes becomes TSE president (Aug); Oct general elections.

2023 — **Jun:** TSE declares Bolsonaro **ineligible for 8 years** (through 2030) — abuse of political power & misuse of media (ambassadors meeting), 5–2.

2023 — **Dec:** STF (ADI 7.261) upholds the TSE's competence to regulate electoral disinformation.

2024 — **Feb:** first-ever AI rules (Res. 23.732/2024) — deepfakes banned, AI-use disclosure required, big-tech liability for non-removal.

2024 — **Oct:** municipal elections. ← *paper's treatment + outcome cycle.*

TSE acts / STF or election. The 2020 and 2024 cycles bracket this period of intense judicial rule-making. Sources: TSE; STF.

What Lawsuits Do — and Who Files Them

Effects on targeted candidates:

- **Candidacy at risk:** court may rule ineligible before election day
- **Reputational cost:** public filing, media coverage, social stigma
- **Resource diversion:** legal defense competes with campaigning
- **Quality signal to voters:** a respondent may be perceived as ethically challenged — or as a victim of political harassment

Who files?
Opposing candidates, parties, *Ministério Público Eleitoral*, any citizen

Treatment Variable

Adversarial competition-relevant cases:

- Filed in the electoral zone before election day
- Substantive challenge (adversarial class and subject — not administrative)
- Includes cases against any candidate type in the zone

Measurement:
$\Delta \log(1 + \text{lawsuits}_m)$
from 2020 to 2024

"Judicialization" here = routing competition *through the courts* (measured by filings) — not

Administrative vs. Adversarial Electoral Litigation

Case type	2020 volume	2024 volume	Included?
Excluded — administrative / mandatory filings			
Candidacy registration (RRC)	∼994k	∼841k	×
Coalition registration (DRAP)	∼137k	∼118k	×
Other administrative classes	various	various	×
Included — adversarial competition-relevant cases			
Eligibility challenges			✓
Abuse of economic/political power			✓
Campaign conduct violations			✓
Propaganda / information offenses			✓
Total adversarial	**∼1.2M**	**∼1.1M**	✓

Pre-election restriction: filing date $<$ election date (2020: Nov 15; 2024: Oct 6). Filter applied at data construction via class and subject-code exclusions. See Appendix D for full case taxonomy.

The Endogeneity Problem

OLS is biased — direction is ambiguous:

Source	Confound
Competition $\rightarrow$ lawsuits	Close races attract more suits $\Rightarrow$ positive bias on competitiveness
Vulnerability $\rightarrow$ lawsuits	Weak incumbents invite challengers to file $\Rightarrow$ negative bias on incumbency
Unobservables	Political culture, lawyer density, party intensity — correlated with both treatment and outcome

Solution: Bartik shift-share IV

Predicted 2024 lawsuits from the municipality's **2020 topic mix** $\times$ **state-level topic growth** — a source of variation the municipality does not drive. Litigation waves are driven by TRE jurisprudence, MPE enforcement priorities, and party legal strategies at the supra-municipal level (Helmke & Staton 2011).

A Theory of Arena-Wide Information Degradation

Core claim:
Adversarial pre-election litigation degrades the *informativeness* of the electoral environment. Voters cannot readily separate a meritorious challenge from strategic harassment, so additional litigation raises *noise*, not signal.

One mechanism, differentiated margins:

- **Candidates:** litigation raises the fixed cost of running (Besley & Coate 1997) $\Rightarrow$ marginal entry deterrence
- **Voters:** unable to update toward any candidate $\Rightarrow$ withdrawal from candidate choice
- **Competition:** no *systematic* reallocation across candidates $\Rightarrow$ competition measures are precise nulls

Why a blank vote, not abstention?

Brazil has **compulsory voting**. Exit (abstention) is penalised, so the disaffected voter's available margin is to appear at the polls and cast a **blank** ballot. The theory therefore predicts disaffection surfacing as blank votes with *turnout unchanged* — exactly the observed pattern (Power & Roberts 1995).

Testable predictions

1. Blank rate $\uparrow$; turnout ≈ 0
2. Candidate count $\downarrow$ (quantity, not selection)
3. Competition outcomes ≈ 0 (no reallocation)
4. Effect strongest where no incumbent anchors voter beliefs (open seats)

Building the Dataset

LAWSUITS

Source: TSE

adversarial pre-election cases by subject code

CANDIDATES

Source: TSE

registration, vote totals, demographics

VOTERS

Source: TSE

turnout, blank, null, valid votes by zone

first round only

CONTEXT

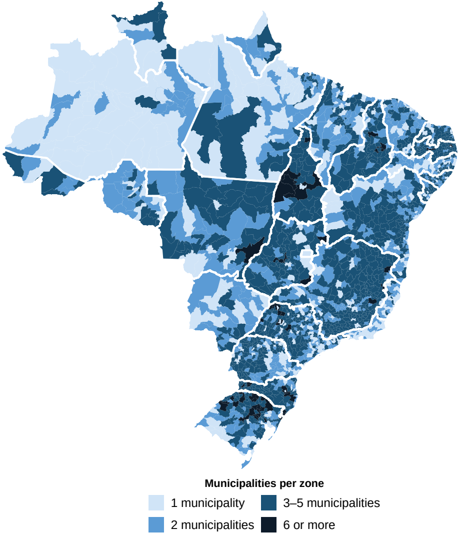
Source: IBGE

population, income, urbanization

Final municipal dataset

$N = 5{,}560$ · first-differenced Δ 2020→2024 · SE clustered by 2,187 electoral zones

Electoral Zones vs. Municipalities



Sample coverage

Municipalities	5,571
Electoral zones	2,190
States (UF)	27
Avg. munis per zone	2.5

Zone size distribution

Munis per zone	Zones	Share
1	1,001	38%
2	609	23%
3–5	965	37%
6 or more	63	2%

Outcome Families

All outcomes are first-differences Δ (2024 – 2020). First round only.

Electoral Competition

- Δ Runner-up vote share
- Δ Margin (winner – runner-up)
- Δ Winner vote share
- Δ P(winner > 50%)
- Δ Log n candidates

Voter Behavior

- Δ Turnout rate
- Δ Null vote rate
- Δ **Blank vote rate** $\leftarrow$ primary
- Δ Valid vote rate

Composition & Entry

- Δ Female vote share
- Δ Nonwhite vote share
- Δ New-candidate vote share
- Δ Incumbent vote share
- Δ Winner is female
- Δ Winner is new entrant
- Δ New entrant pool share
- Δ Serial challenger share
- Δ Cross-cycle returner share

Scale of the Phenomenon — Lawsuits and Voters

Electoral Lawsuits

	2020	2024
All lawsuits (total)	2.70M	2.33M
of which adversarial	1.23M	1.06M
adversarial share	45.6%	45.5%
Municipalities exposed	4,045	4,040
share of total	72.6%	72.5%
Intensity ratios		
All lawsuits / candidate	5.0	5.2
Adversarial / candidate	2.3	2.4
1 adversarial per	121 voters	148 voters

2.3 adversarial lawsuits per candidate on average — litigation is pervasive, not exceptional. Adversarial share is stable (≈46%) across cycles. All intensity ratios **rose** from 2020 to 2024: more lawsuits per candidate (+4%) and one adversarial lawsuit per fewer voters (+22%). Total volume fell −14% — the variation the instrument exploits.

Electorate and Voting Behaviour

	2020	2024
Registered voters	149.4M	156.5M
Voted	**114.8M**	**122.5M**
turnout rate	76.8%	78.3%
Did not vote	**34.6M**	**34.0M**
abstention rate	23.2%	21.7%
Among those who voted		
Blank votes	3.9M (3.4%)	3.5M (2.8%)
Null votes	7.3M (6.3%)	5.3M (4.4%)
Valid votes	103.6M (90.2%)	113.7M (92.8%)

Compulsory voting anchors turnout above 76%. **Blank and null votes** are the residual margin — ≈10% of voters in 2020, falling to ≈7% in 2024. The blank-vote channel is the paper's primary identified effect (+1.3 pp, $p = 0.018$).

Summary Statistics

Variable	Mean	SD	Min	Max
Treatment and instrument				
$\Delta\log(1+\text{adversarial lawsuits})$	−0.107	0.434	−7.56	5.46
Bartik IV (Z_m^{B}, adversarial filter)	−0.121	0.078	−0.62	0.14
Outcome variables				
Δ Runner-up vote share	−0.012	0.158	−0.500	0.499
Δ Margin (winner−runner-up)	0.060	0.325	−1.000	1.000
Δ Winner vote share	0.048	0.192	−1.000	0.718
Δ Blank vote rate	0.001	0.025	−0.069	0.102
Δ Turnout rate	0.010	0.031	−0.158	0.149
Pre-determined controls				
Log population (2010)	9.41	1.15	6.69	16.24
Urban share (2010)	0.638	0.220	0.042	1.000
Log income per capita (2010)	6.231	0.472	4.866	7.702
Margin 2016 (pp)	18.5	21.2	0.0	100.0

$N=5{,}560$ municipalities. Δ blank rate min/max: 1st/99th percentile.

Candidate and Elected Composition (2020 and 2024)

N ≈ 5,568–5,569 municipalities. Weighted by total candidates per municipality.

	Executive (Prefeito)		**Legislative (Vereador)**	
	2020	2024	2020	2024
Candidate pool				
Female share	13.4%	15.3%	34.8%	35.4%
Nonwhite share	36.1%	35.8%	51.3%	53.0%
Higher education	61.4%	63.8%	26.9%	31.1%
Mean age (years)	50.3	50.9	45.7	47.2
Elected pool				
Female share	12.7%	13.2%	14.9%	17.1%
Nonwhite share	32.7%	33.2%	45.0%	45.5%
Higher education	63.6%	66.4%	41.0%	44.9%
Votes cast for female candidates (executive only)				
Female vote share	13.1%	14.6%	—	—

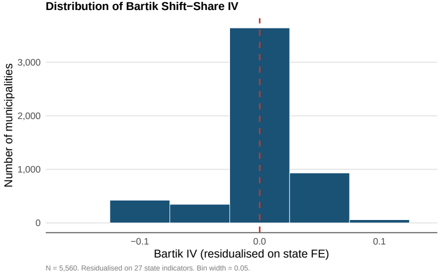
Representation gaps

Executive: entry is the bottleneck, not selection — female candidates (∼14%) and female mayors (∼13%) are nearly equal. No gender quota applies to mayoral races.

Between-cycle change

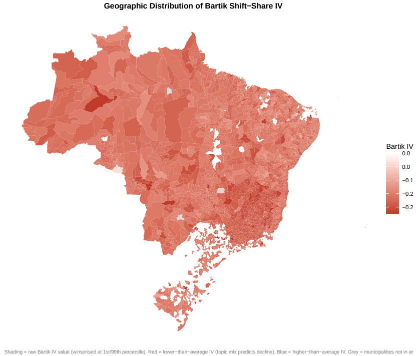
Composition is **stable** (+1–2 pp). Women are 15–16% of mayors; nonwhite candidates are 53% of the legislative field but only 45% of elected councils.

Distribution of Bartik IV



Residualised on state FE. Mean $= -0.121$, SD $= 0.078$.

Geographic distribution



Raw Bartik IV by municipality. Blue $=$ above-average predicted exposure; red $=$ below-average.

Key features: Treatment declined on average 2020→2024 (mean $\Delta = -0.107$). Most variation is cross-sectional: municipalities with a heavier campaign-conduct and information-environment mix in 2020 faced higher predicted litigation waves by 2024. Larger, higher-turnout municipalities had more diverse 2020 portfolios (see balance table, Appendix F).

In 2020, each municipality has a topic mix:
Some had mostly candidate eligibility challenges; others more campaign conduct or propaganda violations.

By 2024, litigation waves diffused unevenly across topics.
Waves spread through TRE jurisprudence, coordinated party strategies, and MPE enforcement priorities — state-wide forces, not municipality-specific.

The instrument:
Z_m^{B} = municipality's *exposure* to these waves, predicted from its 2020 topic mix $\times$ state-level topic growth.

Key Design Logic

1. **Shares**: 2020 topic mix — pre-determined, before 2024 politics materialize
2. **Shifts**: state-level growth driven by institutional diffusion, not any single municipality
3. **Product**: predicted exposure orthogonal to local 2024 outcomes

Template

Ash, Morelli & Vannoni (2025, JPE): same Bartik design applied to legislative output. Litigation topics play the role of legal subject matter.

$$Z^{\mathrm{B}}_m = \sum_{k \notin \mathcal{A}} s_{m,k,2020} \cdot g_{\mathrm{UF}(m),\,k,\,-m}$$

$s_{m,k,2020}$	Municipality m's share of topic k among adversarial lawsuits in 2020; re-normalized so that $\sum_k s_{m,k,2020} = 1$ per municipality
$g_{\mathrm{UF},k,-m}$	Leave-one-out log growth of topic k in state $\mathrm{UF}(m)$, 2020 to 2024: $g = \log\left(\frac{T_{\mathrm{UF},k,2024} - C_{m,k,2024}}{T_{\mathrm{UF},k,2020} - C_{m,k,2020}}\right)$
$\mathcal{A}$	Set of excluded administrative/procedural classes and subjects (incl. RRC, DRAP, mandatory class codes)
LOO	Municipality m excluded from its own state shift to prevent mechanical correlation

Single preferred specification. The adversarial filter is applied *at the data construction stage* via class- and subject-code exclusions, not post-hoc topic removal. This produces one instrument ($F \approx 19.6$) used throughout.

Exclusion Restriction

Formal condition:
$E[Z_m^{\mathrm{B}} \cdot u_m \mid \mathbf{X}_m, \delta_{\mathrm{UF}}] = 0$

GPS (2020) — share exogeneity:
The 2020 litigation portfolio was determined by then-prevailing political dynamics — two election cycles before 2024 outcomes materialize. Conditional on state FE and 2020 controls, shares are as-good-as-random.

BHJ (2022) — shift exogeneity:
State-level topic growth reflects TRE jurisprudence, MPE priorities, and party legal strategies common to a state — not driven by individual municipality dynamics.

Potential Threats

1. **Large-muni dominance**: m drives state shift
   → LOO construction; most munis are small
2. **Share persistence**: 2020 shares correlated with unobserved 2024 dynamics
   → state FE + 2020 controls; balance table (App. F)
3. **Network spillovers**: waves correlated with state events
   → state FE absorbs common shocks
4. $\mathbf{K}_{\mathrm{eff}} \approx 3$: BHJ asymptotics weak
   → GPS approach primary; exposure-robust SEs reported

What the IV Identifies — Interpreting the Estimand

The 2SLS coefficient is **not** a general "effect of judicialization." It is a **weighted LATE**:

- **Weights** (α_k) concentrate on the topics that diffuse at the state level — topics with large state-level growth waves drive the identification
- **Compliers** are municipalities whose litigation responds to those state-wide waves
- The estimate speaks to *these* kinds of adversarial litigation, in *responsive* municipalities — not to all litigation everywhere

Why this matters for framing

Reporting the LATE honestly disciplines the interpretation: the blank-vote effect is the causal response of *state-diffusing adversarial litigation* among complier municipalities. Magnitudes should be read as complier effects, which IV typically scales up relative to a population average.

Exposure-Robust Standard Errors (BHJ/AKM)

Outcome	$\hat{\tau}$	SE_{conv}	SE_{BHJ}	Ratio
Δ Winner majority	+0.140	0.090	2.062	22.9×
Δ Margin	+0.062	0.043	0.787	18.3×
Δ Winner share	+0.031	0.025	0.546	21.8×
Δ Blank rate	+0.013	0.005	0.055	10.4×

$K_{eff} = 3.0$ (Rotemberg). Source: `exposure_robust_se.csv`.

Implication

With only $K_{eff} \approx 3$ topics driving identification, BHJ/AKM SEs are **10–23**× the conventional SEs. Under exposure-robust inference **no outcome is statistically significant** — including the blank rate ($t_{BHJ} = 0.23$). This reflects the low effective topic count, not a failure of the instrument per se.

What this means

The tF correction adjusts for *weak instruments*; BHJ/AKM adjusts for *estimated shifts* (few effective topics). Both matter here because $K_{eff} = 3$.

GPS approach is primary: with many small municipalities, the GPS share-exogeneity argument holds even at low K_{eff}. Conventional SEs clustered by zone are appropriate under GPS, and the tF-corrected result holds.

Honest disclosure: under the more conservative BHJ criterion, the blank-rate finding is suggestive, not confirmed. Improving K_{eff} is a key agenda item.

Regression Equations

First stage (OLS):

$$\underbrace{\Delta \log(1+\ell_m)}_{\text{endogenous}} \;=\; \alpha + \beta\, Z^{\mathrm{B}}_m + \mathbf{X}'_m\gamma + \delta_{\mathsf{UF}(m)} + \varepsilon_m$$

Second stage (2SLS):

$$\Delta Y_m \;=\; \alpha + \tau\, \Delta \widehat{\log(1+\ell_m)} + \mathbf{X}'_m\gamma + \delta_{\mathsf{UF}(m)} + u_m$$

- ℓ_m = adversarial competition-relevant pre-election lawsuits (respondent is mayoral candidate)
- Δ = first difference 2024−2020
- $\mathbf{X}_m$ = 7 baseline pre-determined controls
- $\delta_{\mathsf{UF}(m)}$ = state fixed effects (27 units)
- SE clustered by principal electoral zone (2,187 clusters)
- tF critical values (Lee et al. 2022) reported for weak-instrument-robust inference given $F = 19.6 < 23.1$

Specifications

Specification	Sample	N
Baseline	All municipalities	5,560
Single-zone munis	One zone only	5,371
Extended controls	All municipalities	5,560
Open seat	2020 winner term-limited	2,571
Contested seat	Incumbent can run	2,989
Broader treatment	All municipalities	5,560

Controls:

Baseline (7): Census 2010 (log pop, urban share, log income pc) + margin 2016 + 2020 baseline (log valid votes, margin, log candidates)

Extended (+7): ENP 2016 + candidate composition 2020 (gender, ethnicity, education) + party count

What each spec tests:

- **Baseline** — primary estimates; state FE + 7 controls
- **Single-zone** — institutional check: municipalities served by one zone avoid zone-to-municipality aggregation ambiguity
- **Extended controls** — controls robustness; adds candidate pool composition and party fragmentation
- **Open/contested seat** — mechanism test: blank-vote effect should concentrate where no incumbent anchors voter beliefs
- **Broader treatment** — filter robustness; adds log non-RRC lawsuits as a control to verify adversarial measure drives results

First-Stage Results

Specification	Coef.	SE	t	F-stat	tF_{cv}	N
Baseline	1.791	0.405	4.43	**19.6**	2.18	5,560
Single-zone munis	1.801	0.422	4.27	**18.2**	2.23	5,371
Extended controls	1.799	0.406	4.44	**19.7**	2.18	5,560
Broader treatment	1.792	0.405	4.43	**19.6**	2.18	5,560

SE clustered by principal electoral zone. $F = (t\text{-stat})^2$. tF_{cv} from Lee et al. (2022) Table 1.

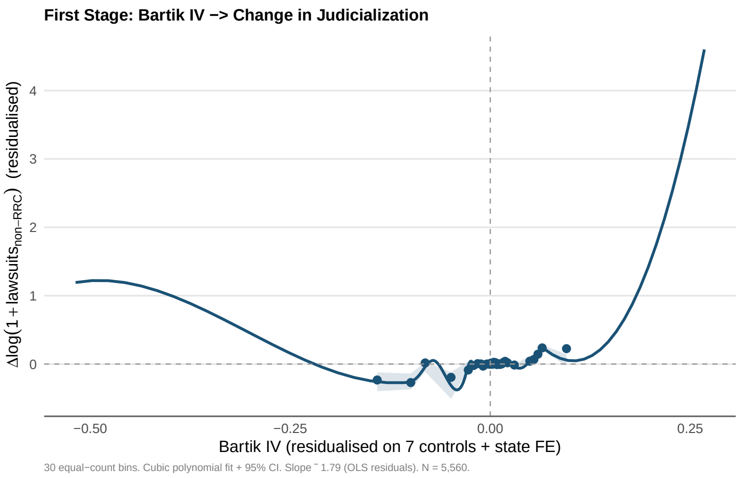
Instrument Strength

$F = 19.6$, stable across all four specifications. Stock–Yogo 10% size distortion critical value $\approx$10; tF threshold for 5% inference = 23.1. We apply tF correction throughout.

LIML Check

For a just-identified model ($K = 1$ instrument), LIML $\equiv$ 2SLS exactly. Confirmed: max coefficient divergence = 0.0000 across all outcomes. No many-instrument or weak-instrument bias.

First Stage — Binscatter



Electoral Competition — Baseline Specification

Adversarial instrument ($F = 19.6$). $N = 5{,}560$, clusters $= 2{,}187$. State FE + 7 controls.

Outcome	Coef.	SE	p	t	tF reject?
Δ Runner-up vote share	−0.031	0.022	0.157	−1.42	×
Δ Margin (W−RU)	+0.062	0.043	0.151	+1.44	×
Δ Winner vote share	+0.031	0.025	0.213	+1.25	×
Δ Winner majority (P50)	+0.140	0.090	0.122	+1.55	×
Δ Log n candidates	−0.066	0.039	0.092	−1.69	×

Directional pattern (winner consolidation)

All competition outcomes point toward winner consolidation: higher majority probability, wider margin, fewer candidates. Direction is consistent across all 4 main specifications — though these share the instrument and sample, so consistency is suggestive, not independent confirmation. **None survives tF correction** ($tF_{cv} = 2.18$; max $|t| = 1.69$).

Candidate pool: marginal signal

Log candidates (−6.6%, $p = 0.092$) is the only outcome approaching conventional significance. Consistent with Besley & Coate (1997): adversarial challenges raise the fixed cost of candidacy, deterring entry at the margin. Replicated in legislative analysis (−7.5%, $p = 0.066$).

Electoral Competition — Robustness Across Specifications

Outcome: Δ winner majority. All specifications, adversarial instrument.

Specification	Coef.	SE	p	F	N
Baseline	+0.140	0.090	0.122	19.6	5,560
Single-zone munis	+0.131	0.093	0.160	18.2	5,371
Extended controls	+0.136	0.086	0.116	19.7	5,560
Broader treatment	+0.140	0.090	0.122	19.6	5,560

- Direction and magnitude are remarkably stable: coefficient ranges +0.131 to +0.140 across all specifications
- The result is not driven by control set, sample, or the non-adversarial baseline
- The precision gap (not reaching $p < 0.05$) is the fundamental limitation; a larger panel or stronger instrument would be needed to confirm
- **Pre-period placebo** (2016 $\rightarrow$ 2020 outcomes): the ideal test would regress 2016–2020 outcome changes on the 2020–2024 Bartik IV. Requires the 2016 lawsuit panel, which is not yet downloaded. The test would be partial even then (ideal shares would be from 2016, not 2020). *Pending: TSE processual data 2016.*

Voter Behavior — Baseline Specification

$N = 5{,}560$, state FE + 7 controls. Rates denominated on registered voters.

Outcome	Coef.	SE	p	t	tF reject?
Δ Turnout rate	−0.006	0.005	0.185	−1.33	×
Δ Null vote rate	+0.007	0.006	0.220	+1.23	×
Δ Blank vote rate	**+0.013**	**0.005**	**0.018**	**+2.36**	✓
Δ Valid vote rate	**−0.026**	**0.013**	**0.038**	**−2.07**	×

Main finding: blank vote rate

2SLS coefficient: $+0.013$ per unit of $\Delta \log(1 + \text{lawsuits})$. A municipality whose litigation doubled ($\log \approx 0.69$) would see blank rate rise by $\approx$0.9 pp — from a base of 1.5%, a $\approx$60% relative increase among compliers.

Effect survives tF correction ($|t| = 2.36 > \text{tF}_{\text{cv}} = 2.18$). Valid-vote rate ($p = 0.038$) is the accounting mirror and does *not* survive tF. **Blank rate is the primary identified effect; robustness to exposure-robust SEs discussed in Appendix.**

Precise nulls

Turnout ($p = 0.185$) and null rate ($p = 0.220$) are precise nulls. Adversarial judicial challenges do *not* demobilize the electorate, contrary to negative advertising effects (Ansolabehere & Iyengar 1995). The behavioral channel runs through expressive abstention, not non-participation.

Blank Vote Rate — Interpretation

What blank votes signal:
In Brazil, a blank vote (*voto em branco*) is a deliberate choice to reject all listed candidates without spoiling the ballot. It requires specific action (pressing a button or writing 'branco').

Mechanism:
Adversarial judicial challenges are publicly filed records that generate information about candidates' alleged conduct. Voters exposed to more diverse and intense judicial activity may face:

- Increased uncertainty about all candidates' quality
- Negative updating about the entire political class (*not* specific to a targeted candidate)
- Expressive dissatisfaction with the judicially-contested electoral environment

Literature anchor

Ferraz & Finan (2008): publicly released audit information changes electoral outcomes in Brazil. Judicial filings are analogous: they are publicly available before election day and carry information about alleged wrongdoing.

Key difference: audits target specific corrupt incumbents (directional information); adversarial judicial waves are diffuse across all candidates (ambiguity-generating). This explains the blank vote outcome rather than an incumbent vote share effect.

Robustness

Effect stable: coefficient 0.012–0.013 across baseline, single-zone, and extended-controls specifications.

Interpreting the Magnitude

Scale of the effect

- Coefficient: $+0.013$ per unit of $\Delta \log(1 + \text{lawsuits})$
- A doubling of lawsuit volume ($\Delta \log \approx 0.69$) implies $\approx$0.9 pp rise in blank rate
- Mean blank rate 2020 $\approx 1.5\%$: a doubling-induced rise $= +60\%$ relative increase for compliers
- Cross-municipal SD of Δblank rate $= 0.025$; effect of 1 SD of endogenous ≈ 0.006 pp — small in distribution terms, but instrument-driven variation is concentrated in large shocks

LATE interpretation

The estimate is a complier LATE: it identifies municipalities pushed into higher adversarial litigation by state-wide institutional waves. Complier

Benchmark: Ferraz & Finan (2008)

Publicly released corruption audits increased incumbent loss probability by 7–17 pp in Brazil. **Channel difference:** audits provide *directed* information about a specific incumbent's corruption; adversarial litigation provides *diffuse* signals across all candidates. Directed signals change vote shares; diffuse signals increase blank votes. The channels are complementary — both consistent with information-driven voter responses.

Caveat

Under BHJ/AKM exposure-robust SEs (see previous slide), the blank rate effect does not survive at conventional significance levels. The magnitude interpretation assumes tF-corrected inference is appropriate under GPS identification

Open-Seat Heterogeneity — Blank Vote Rate

Open seat (46% of municipalities, $N = 2{,}571$):
2020 winner served two consecutive terms $\Rightarrow$ constitutionally term-limited in 2024. No incumbent on the ballot. Exogenous to 2024 litigation activity.

Contested seat (54%, $N = 2{,}989$):
2020 winner's first term. Incumbent can seek reelection. Voters have a focal candidate (support or oppose the incumbent).

Sample	Coef.	SE	p	F	tF?
All	+0.013	0.005	0.018	19.6	✓
Open seat	**+0.030**	0.014	0.028	10.6	×*
Contested	+0.003	0.005	0.546	15.3	×

*Conventional $p = 0.028$ but $|t| = 2.20 < \text{tF}_{cv} = 2.78$ ($F = 10.6$).

Mechanism

The aggregate blank-rate effect runs almost entirely through **open-seat municipalities**.

Why? When no incumbent is running, voters lack a focal candidate. Adversarial judicial activity creates ambiguity about *all* contenders simultaneously — heightening the attractiveness of a blank vote.

When the incumbent runs, voters already have a valence benchmark (vote for or against the mayor). Legal noise is absorbed: the contested-seat effect is a structural zero.

Caveat: weakened instrument

Open-seat subgroup: $F = 10.6$ (vs. 19.6 full sample). Smaller municipalities likely to have longer-serving incumbents but also less electoral litigation variation. The tF correction is therefore stricter ($\text{tF}_{cv} = 2.78$); effect survives conventional but not weak-IV-robust inference.

Additional Heterogeneity: Competitive Races

Does the blank-rate effect concentrate in close races?

Hypothesis: judicial noise has higher marginal impact when voter uncertainty is already high (tight race). Split at the median 2020 margin (13.4 pp).

Sample	Coef.	SE	p	F
Δ *Blank rate*				
Competitive	+0.006	0.004	0.139	175.9
Uncompetitive	+0.024	0.013	0.068	73.3
Δ *Log candidates*				
Competitive	−0.071	0.043	0.096	175.9
Uncompetitive	−0.082	0.075	0.275	73.3

Split at median margin (0.134). Baseline spec.

Result: opposite to prior hypothesis

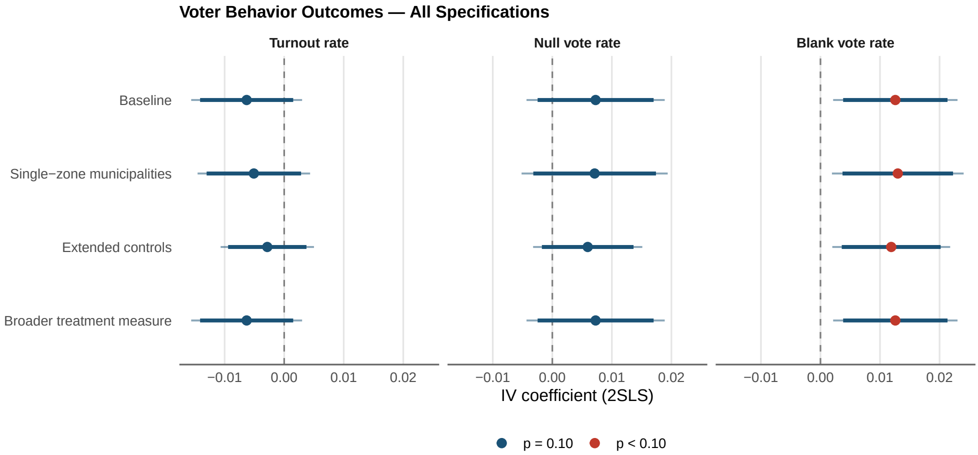
The blank-rate effect is **larger in uncompetitive races** (+2.4 pp, $p = 0.068$) than competitive ones (+0.6 pp, $p = 0.139$). This mirrors the open-seat finding: when voters have less of an anchor (either no incumbent or a dominant candidate), judicial noise translates more readily into blank votes.

Candidate deterrence is directionally similar across groups but imprecise in the uncompetitive subsample.

Pending heterogeneity

- **Information environment:** internet penetration / broadband (IBGE) — data available, not yet merged
- **Legal capacity:** OAB lawyer counts by municipality — data available, not yet merged
- **TRE enforcement:** state-level variation —

Voter Behavior — Forest Plot (All Specs)



Thick bars: 90% CI. Thin bars: 95% CI. SE clustered by principal electoral zone.

Candidate Pool and Composition — Baseline

Adversarial instrument, baseline specification. $N = 5{,}560$, clusters $= 2{,}187$.

Outcome	Coef.	SE	p	tF reject?
Δ Log n candidates (exec.)	−0.066	0.039	0.092	×
Δ New-cand. vote share (exec.)	−0.010	0.050	0.848	
Δ Incumbent vote share (exec.)	+0.010	0.049	0.844	
Δ Female vote share	−0.093	0.048	0.052	×
Δ Nonwhite vote share	−0.039	0.042	0.348	
Δ Winner is female	−0.123	0.063	0.052	×
Δ Winner is new entrant	−0.002	0.073	0.982	

Entry deterrence (not composition shift)

The candidate count reduction (−6.6%) is directional, not significant. But it replicates across executive and legislative arenas (next slide), lending it more credibility. New-entrant and incumbent vote shares are both near zero — the field shrinks, but without systematic reallocation toward any type.

Gender: directional, not robust

Female vote share (−9.3 pp, $p = 0.052$) and female winner probability (−12.3 pp, $p = 0.052$) both have $|t| = 1.94 < \text{tF}_{\text{cv}} = 2.18$; direction is suggestive but not robust. Nonwhite vote share (−3.9 pp, $p = 0.348$) is a precise null — adversarial litigation does not selectively displace nonwhite candidates.

Candidate Entrant Typology — New Variables

Three entrant types, 4 electoral cycles (2012–2024):

- **Pure new entrant:** no prior mayoral candidacy anywhere in 2012, 2016, or 2020 ($\bar{s}_{2024} = 55\%$)
- **Serial challenger:** ran in the same municipality in 2020, lost, running again ($\bar{s}_{2024} = 13\%$)
- **Cross-cycle returner:** has prior history but sat out 2020 entirely; returning after a cycle gap ($\bar{s}_{2024} = 9\%$)

Question: Does adversarial judicialization reshape *who enters*, not just *how many*?

IV Results: Entry Composition

Outcome	Coef.	SE	p
Δ New entrant share	−0.011	0.043	0.793
Δ Serial challenger	+0.007	0.026	0.802
Δ Cross-cycle ret.	+0.004	0.025	0.861

Baseline spec. N = 5,560, F = 19.6.

Interpretation

All three entrant-type shares are precise nulls. Judicialization does not *selectively screen* entrant types — it reduces the count (−6.6%) without reshaping the pool's experience composition.

Entry deterrence is a **quantity** effect, not a **selection** effect.

Legislative Analysis — City Council Elections (Vereadores)

Same instrument, different electoral arena:

- Brazil's city councils (*Câmaras Municipais*) elect 9–55 *vereadores* depending on population (CF Art. 29-IV)
- Pre-election challenges also target legislative candidates
- **Multi-seat proportional** system — different from the plurality mayoral race

Identification: same municipality-level Bartik IV ($F \approx 19.4$). The shift in adversarial litigation waves applies to the same municipality and electoral environment. Endogenous: $\Delta \log(1 + \text{competition lawsuits})$ at municipality level (same treatment).

Outcome Families

Candidate pool:

- Total candidates, female/nonwhite shares
- New candidates, incumbents, party count

Elected composition:

- Elected female/nonwhite shares
- Higher-education share, mean age
- Incumbent reelected share

Party competition:

- Party count, coalition count

Instrument

$N = 5{,}560$; clusters $= 2{,}187$; $F = 19.4$.

Legislative IV Results

Baseline specification. $N = 5{,}560$, clusters $= 2{,}187$. State FE + 7 controls. $F = 19.4$.

Outcome	Coef.	SE	p	Sig.
Candidate pool				
Δ Log total candidates	−0.075	0.041	0.066	†
Δ Female candidate share	−0.006	0.006	0.370	
Δ Nonwhite candidate share	−0.018	0.019	0.346	
Δ New candidate share	−0.020	0.016	0.217	
Δ Eff. party count (candidates)	−0.264	0.301	0.381	
Elected composition				
Δ Elected female share	+0.013	0.022	0.554	
Δ Elected nonwhite share	−0.027	0.026	0.288	
Δ Elected higher-ed share	+0.006	0.026	0.802	
Δ Incumbent reelected share	+0.006	0.025	0.797	
Party competition				
Δ Party count	−0.260	0.354	0.463	
Δ Coalition count	+0.074	0.066	0.266	

†$p < 0.10$. SE clustered by principal electoral zone.

Executive vs. Legislative: Comparing Effects

Same instrument (F $\approx$ 19–20), two electoral arenas:

Outcome	Executive	Legislative
Δ Log candidates	$-0.066^{\dagger}$	$-0.075^{\dagger}$
Winner consolidation	directional	—
Female share	$-0.093^{\dagger}$	-0.006
Blank rate	$+0.013^{**}$	—
Elected composition	null	null
Party competition	null	null

$^{**}p < 0.05$ (tF corrected); $^{\dagger}p < 0.10$ (uncorrected).

Interpretation

Both arenas: candidate pool contracts (-7%) consistently — entry deterrence channel is the most robust structural finding. Consistent with Besley & Coate (1997).

Executive only: winner consolidation (directional), blank rate (significant), gender effects (borderline) — these run through competition dynamics specific to the single-winner plurality race.

Legislative: multi-seat proportional allocation dampens the competitive displacement effects visible in executive races. No effect on who wins seats or on party composition.

Motivation:
The aggregate Bartik IV combines topics from four litigation families. Do different *types* of adversarial challenges operate through different mechanisms?

Approach (AMV 2025, Section 6):
Construct a *family-specific* Bartik IV for each topic family:

$$Z_m^f = \sum_{k \in f} s_{m,k,2020} \cdot g_{\mathrm{UF},k,-m}$$

Each Z_m^f instruments the *family-specific* endogenous: $\Delta \log(1 + \ell_{m,f})$. This identifies a family-level LATE — different from the aggregate.

Four Litigation Families

Family	F	N	Identified?
Campaign conduct	36.0	3,937	✓
Information env.	19.7	3,861	✓
Abuse of office	7.8	3,311	×
Eligibility/Ballot	8.4	1,772	×

Campaign conduct: campaign propaganda, prohibited conduct, vote-buying

Information environment: media, internet propaganda, right of reply

Eligibility/Ballot access: registration challenges, impugnação

Abuse of office: economic and political power abuses

Family IV Results — Primary Outcomes

Family-specific Bartik IV instruments family-specific endogenous. Baseline specification + 7 controls.

	Δ Runner-up		Δ Margin		Δ Log cand.	
Family (F^{fs})	Coef.	p	Coef.	p	Coef.	p
Campaign ($F = 36$)	−0.007	0.545	+0.018	0.442	−0.004	0.836
Info. env. ($F = 20$)	−0.010	0.511	+0.034	0.250	+0.013	0.665
Abuse of office ($F = 7.8$) and eligibility ($F = 8.4$): weak instrument — estimates unreliable.						

Null across well-identified families

Both campaign-conduct and information-environment sub-instruments produce null effects across all competition outcomes. This is consistent with the aggregate null — the aggregate result is not concealing strong but cancelling effects across family types. Direction is non-systematic (not all positive or all negative), further supporting the null.

Implications

The blank-rate effect and the candidate-pool reduction are not attributable to a single mechanism family with the currently available family instruments. Both campaign conduct and information environment challenges contribute to the aggregate treatment without producing detectable competition effects through their separate channels.

Mechanism: Ruling Out Alternative Explanations

Direct disqualification If lawsuits primarily remove candidates from the ballot (*cassação*), the operative channel is a mechanical reduction in ballot options rather than voter ambiguity. This predicts directional vote-share shifts toward remaining candidates (incumbents gain, challengers lose). **Not observed:** incumbent and new-entrant vote shares are precise nulls ($p > 0.84$). Entry deterrence is a candidate-registration decision, not a disqualification outcome. Requires case-outcome data to test directly.

Campaign resource diversion Targeted candidates divert legal defense resources away from campaigning, mechanically reducing their vote share. Predicts *individual-level* effects concentrated on targeted candidates. Our municipality-level instrument cannot separate this from the arena-wide noise channel. Proxy test (case complexity as cost measure) requires merging partes data — pending.

Directed information vs. diffuse noise If litigation conveys genuine signal about candidate quality, voters should reallocate away from targeted candidates toward untargeted ones. **Data rules this out:** female vote share ($p = 0.052$), new-entrant share ($p = 0.848$), and incumbent share ($p = 0.844$) all show no systematic reallocation. Voters appear to update toward *all* candidates negatively — consistent with noise, not signal.

Robustness: Extended Controls and Broader Lawsuits

14-variable extended control set:
Adds ENP 2016, candidate composition (gender, ethnicity, education), party/coalition count, voter behavior baseline, candidate experience.

Caveat: some extended controls are outcome-adjacent (2020 candidate composition may reflect prior judicialization). Use as sensitivity only.

Broader treatment measure:
Adds $\log(1 + \ell_m^{\text{no-RRC}})$ in 2020 as a covariate — the broader competition lawsuit stock (excluding only RRC). Tests whether the adversarial IV effect is distinct from general pre-existing litigation activity.

Blank rate across control sets

Spec	Coef.	SE	p	F
7 baseline	$+0.013$	0.005	0.018	19.6
14 extended	$+0.012$	0.005	0.022	19.7
Broader ctrl	$+0.013$	0.005	0.018	19.6
Margin (7 base.)	$+0.062$	0.043	0.151	19.6
Margin (14 ext.)	$+0.057$	0.042	0.180	19.7
Margin (broader)	$+0.062$	0.043	0.151	19.6

Blank rate result is stable to control set and to conditioning on broader lawsuit baseline.

Welfare and Normative Implications

Framing 1: Healthy accountability

Lawsuits expose genuine candidate misconduct. Rational voters facing a contentious legal environment become more cautious — blank votes reflect sound democratic skepticism, not manipulation. Under this view, the TSE's enforcement expansion is welfare-improving: it disciplines candidates and informs voters, even if some voters respond with expressed dissatisfaction.

Prediction: high conviction rates, meritorious cases. Deterred candidates are those who would have behaved illegally.

Framing 2: Democratic erosion / lawfare

Strategic litigation manufactures ambiguity without improving candidate quality. Blank votes reflect suppressed valid voter choice. Entry deterrence is an unambiguous welfare cost if deterred candidates are net-positive for democratic representation.

Prediction: high dismissal rates, filer identity is opposing party, effects concentrated in competitive races. Our evidence of larger effects in uncompetitive races weakly favors the accountability framing — but is not conclusive.

Distinguishing the two requires case-outcome data (win/loss rates by topic and filer type) — currently not merged. A high adversarial dismissal rate would strongly favor the lawfare

Compulsory voting The blank-ballot margin is specific to settings where abstention is penalised. In voluntary-voting systems, the theoretically equivalent response is turnout decline — voters disengage by not showing up rather than by casting a blank. Brazil's electorate ($\sim$156M registered) is the world's largest compulsory voting system. Replication in voluntary-voting settings (e.g., Colombia's recent municipal elections) would require a turnout outcome as the primary channel.

Pre-election first-instance jurisdiction Most electoral courts worldwide act *post-election* (cassação after the fact). Brazil's TRE/TSE system is unusual in adjudicating eligibility and conduct *before* election day. The results speak to this pre-electoral filing process. Post-electoral judicial challenges likely operate through different mechanisms (uncertainty about whether the winner will be seated), which our design cannot identify.

Electoral system Blank-rate and candidate-count effects are identified in single-winner plurality (executive) races. The vereadores results (no blank-rate effect, similar candidate deterrence) are consistent with proportional representation dampening the competitive displacement channel. The arena-wide noise mechanism may be system-agnostic; the competitive outcomes are not.

1. **Blank vote rate — main finding:** $+1.3$ pp ($p = 0.018$; survives tF correction). Adversarial judicialization generates voter ambivalence, expressed through deliberate blank votes. *Under compulsory voting, exit (abstention) is penalised — so disaffection surfaces as a blank ballot.* Turnout and null rate: precise nulls. No demobilization.

2. **Candidate pool contraction (marginal):** -6.6% executive ($p = 0.092$); -7.5% legislative ($p = 0.066$). Consistent across arenas — entry deterrence is the most robust structural channel.

3. **Winner consolidation (directional):** All competition outcomes favor the winner: higher majority, wider margin. Direction stable across all 4 main specifications (which share

Main Takeaway

Credibly identified estimates from a Bartik shift-share IV ($F = 19.6$, tF-corrected inference).

The primary identified effect — a LATE for municipalities pushed into more litigation of the state-diffusing types — is voter ambivalence: a significant rise in blank votes, rather than competitive displacement or mobilization. Under compulsory voting, this is how disaffection surfaces: a blank ballot, not abstention.

Entry deterrence is the most consistent structural finding across arenas.

Contributions, Open Questions, and Next Steps

Contributions:

- First Bartik IV applied to electoral judicialization; blueprint for causal identification of litigation effects in political economy
- Full GPS/BHJ/AKM compliance; tF correction applied
- Systematic administrative-vs-adversarial distinction: a measurement contribution to the electoral law literature
- Cross-arena comparison (executive vs. legislative): entry deterrence replicates; competition effects arena-specific

Open questions requiring new data:

- **Pre-period placebo** (2016 $\rightarrow$ 2020): needs TSE processual data 2016. Ideal test: regress 2016–2020 outcome changes on a 2016-share Bartik IV. Would rule out pre-existing trends

Pending analysis

- Romano–Wolf multiple testing correction across outcome families
- Concavity analysis: zero-share vs. positive-share municipalities (intensive vs. extensive margin)
- ✓ Open-seat heterogeneity: done (blank rate 3$\times$ larger in open seats)
- ✓ Entry typology: done (quantity effect, not selection)
- Female candidate heterogeneity: candidate-level analysis of gender differential exposure

Literature Connections

Methods: GPS (2020); BHJ (2022); AKM

Adversarial instrument (`bartik_iv_2020_2024`). Single specification.

Adversarial filter applied at data construction stage (class and subject-code exclusions from TSE taxonomy; mandatory administrative filings excluded). One instrument, one endogenous variable.

Specification	Coef.	SE	t	F-stat	tF_{cv}	N
Baseline	1.791	0.405	4.43	**19.6**	2.18	5,560
Single-zone munis	1.801	0.422	4.27	**18.2**	2.23	5,371
Extended controls	1.799	0.406	4.44	**19.7**	2.18	5,560
Broader treatment	1.792	0.405	4.43	**19.6**	2.18	5,560

SE clustered by principal electoral zone. $F = (t\text{-stat})^2$.
tF_{cv}: weak-IV-robust critical value (Lee et al. 2022, Table 1), 5% level.

Appendix B: Full IV Table — Competition and Voter Behavior I

Adversarial instrument. Baseline specification. N = 5,560, clusters = 2,187.

Outcome	Coef.	SE	t	p	tF*?
Competition					
Δ Runner-up	−0.031	0.022	−1.42	0.157	×
Δ Margin	+0.062	0.043	+1.44	0.151	×
Δ Winner vote	+0.031	0.025	+1.25	0.213	×
Δ Majority	+0.140	0.090	+1.55	0.122	×
Δ Log candidates	−0.066	0.039	−1.69	0.092	×
Voter behavior					
Δ Turnout rate	−0.006	0.005	−1.33	0.185	×
Δ Null rate	+0.007	0.006	+1.23	0.220	×
Δ Blank rate	**+0.013**	**0.005**	**+2.36**	**0.018**	✓
Δ Valid vote rate	−0.026	0.013	−2.07	0.038	×

*$\text{tF}_{\text{cv}} = 2.18$ (F = 19.6); ✓ = $|t| > 2.18$. SE clustered by principal electoral zone.

Appendix C: Full IV Table — Composition I

Adversarial instrument. Baseline specification. N = 5,560.

Outcome	Coef.	SE	t	p
Δ Female vote share	−0.093	0.048	−1.94	0.052
Δ Nonwhite vote share	−0.039	0.042	−0.94	0.348
Δ New cand. vote share	−0.010	0.050	−0.19	0.848
Δ Incumbent vote share	+0.010	0.049	+0.20	0.844
Δ Winner is female	−0.123	0.063	−1.94	0.052
Δ Winner is new entrant	−0.002	0.073	−0.02	0.982

SE clustered by principal electoral zone. $tF_{cv} = 2.18$; no composition outcome survives tF correction.

Competition-relevant adversarial case criteria:

1. Filed before election day in the electoral zone
2. Case class is not in the administrative exclusion list (DROP_CLASSES)
3. Case subject is not in the procedural exclusion list (DROP_SUBJECTS)

Examples of included subjects:
Abuso do Poder Econômico, Abuso do Poder Político, Captação Ilícita de Sufrágio, Propaganda Eleitoral Irregular, Conduta Vedada a Agente Público, Uso Indevido dos Meios de Comunicação, Impugnação de Registro de Candidatura, Falsificação de Documento

Case Volume (2020 / 2024)

Category	2020	2024
Mandatory registration (RRC)	∼994k	∼841k
Coalition registration (DRAP)	∼137k	∼118k
Adversarial competition cases	∼1.2M	∼1.1M

Filter logic

The adversarial filter excludes entire administrative *classes* (mandatory processing) as well as specific *subjects* that are procedurally rather than substantively adversarial. This is stricter than simply removing RRC and DRAP topic entries.

Appendix F: Balance Table I

Adversarial instrument regressed on each covariate. State FE, SE clustered by zone.

Covariate	Coef.	SE	p
Log population (2010)	-0.543	0.256	0.034
Urban share (2010)	-0.024	0.028	0.385
Log income per capita (2010)	$+0.164$	0.049	0.001
Margin 2016	$+0.089$	0.041	0.030
ENP 2016	-0.256	0.142	0.072
Turnout rate 2020	$+0.040$	0.012	0.001
Female candidate share 2020	-0.075	0.046	0.101
Nonwhite candidate share 2020	-0.050	0.067	0.458
Log n candidates 2020	-0.127	0.084	0.129

4 of 9 covariates $p < 0.05$. Correlations reflect that larger, richer, higher-turnout municipalities had more diverse 2020 litigation portfolios. All significant covariates are included in the baseline control set.

Interpretation: Non-random share exposure is expected when municipalities differ systematically. The identification claim is

Appendix F: Balance Table II

that these differences are absorbed by the pre-determined controls and state FE. The GPS balance tests (gps_balance_tests.csv) run more detailed regressions per top topic.

Appendix G: Family IV — First Stage by Family

Each family IV instruments its family-specific endogenous ($\Delta \log(1 + \ell_{m,f})$). Baseline spec.

Family	Endog. coef.	SE	F	N	Identified?
Campaign conduct	+1.61	0.27	36.0	3,937	✓
Information env.	+2.29	0.52	19.7	3,861	✓
Abuse of office	−2.54	0.91	7.8	3,311	×
Eligibility/Ballot	−0.20	0.07	8.4	1,772	×

Notes: N varies by family because family-IV is defined only for municipalities with nonzero family exposure ($s_{mk,2020} > 0$ for topic k in family f). Negative coefficients for abuse and eligibility families indicate the Bartik IV for those families is negatively associated with own-family lawsuit growth — consistent with diffuse spillover dynamics. These families should not be interpreted.

Abuse of office and **eligibility/ballot** families have F $<$ 10; weak-instrument concerns preclude reliable 2SLS inference. **Campaign conduct** (F = 36) and **information environment** (F = 20) are the identified families.